

Sonar Water-Level Meter



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Three-fourths of the earth is water, out of which 97 per cent is saline (in oceans, seas and groundwater). The remaining 2.5 per cent to 2.75 per cent is fresh water, out of which 1.75 per cent to two per cent is frozen in glaciers, ice and snow. Fresh groundwater and soil moisture constitutes only 0.7 per cent to 0.8 per cent. Less than 0.01 per cent, available as surface water in lakes, swamps and rivers, is available to us as drinking water. It is therefore imperative that systems are in place for managing this precious and scarce resource.

Water-level indicators for water-storage tanks are mostly mechanical contraptions. Steel wires, pulleys and rails get rusted as these are in continuous contact with water and air, resulting in unreliable operation. Contactless sonar-level metres are there but their high cost is a deterrent.

In this article, a sonar contactless, wireless water-level indicator, which can be fabricated for less than ₹ 2500, has been described. It is capable of measuring water levels up to four metres with an accuracy of less than 5mm.

Circuit and working

The water-level indicator has two units: transmitter unit and receiver unit. The circuit diagram of the transmitter unit is shown in Fig. 1. It is built around ATmega328P microcontroller (MCU) (IC1) with Arduino Uno bootloader, sonar sensor HC-SR04 connected at CON1, 433MHz

transmitter (TX1), voltage regulator 7805 (IC2) and a few other components. A long spiral antenna is used

for enhancement of its range.

Attach the contactless sonar sensor at a strategic location on the tank

so that it can always get the reflected signal from the water surface. The best place would be at the centre of a circular tank's lid on top, or at the intersection of the diagonals of a rectangular tank's top. The calculated tank level will then be transmitted with a coded authorisation to the receiver unit.

An ultrasonic ranging module

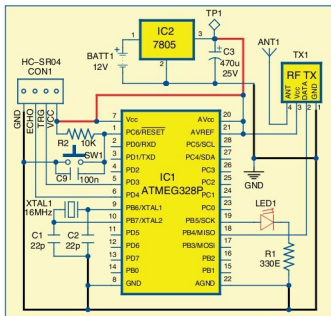


Fig. 1: Circuit diagram of the transmitter unit

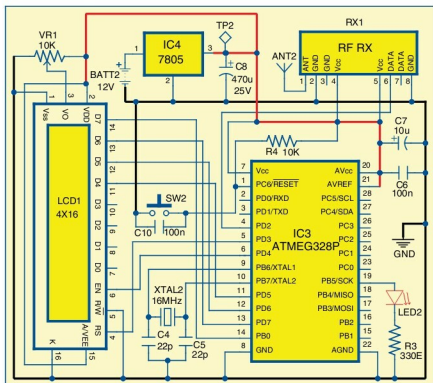


Fig. 2: Circuit diagram of the receiver unit

Test Points

Test point	Details
TP1	+5.0V
TP2	+5.0V
(Voltages measured with respect to ground)	